

A Minimax Lower Bound for Sobolev Nonparametric Regression

1 Setup and preliminaries

Notation. We write $[N] := \{1, \dots, N\}$. For a function u on $[0, 1]^d$, $\|u\|_{L^2}^2 := \int_{[0, 1]^d} u(x)^2 dx$ denotes its squared L^2 norm, and for an integer $s \geq 1$ its squared Sobolev norm is

$$\|u\|_{W_2^s}^2 := \sum_{|\alpha| \leq s} \|\partial^\alpha u\|_{L^2}^2, \quad (1)$$

the sum running over multi-indices $\alpha \in \mathbb{N}^d$ of order $|\alpha| \leq s$, with ∂^α the corresponding weak derivative. The Sobolev class of interest is the unit ball $\mathcal{W} := \{f \in W_2^s([0, 1]^d) : \|f\|_{W_2^s} \leq 1\}$. For $\tau, \tau' \in \{0, 1\}^{M_0}$, $\rho_H(\tau, \tau') := \sum_j \mathbf{1}\{\tau_j \neq \tau'_j\}$ is the Hamming distance and $\|\tau\|_0 := \sum_j \mathbf{1}\{\tau_j \neq 0\}$ the Hamming weight. Throughout, $c, C, c_1, C_1, \dots$ denote universal positive constants in the sense that they may depend only on the fixed quantities (s, d, g, σ^2) and on no other problem parameter (in particular not on the sample size n or the resolution m); their values may change from line to line.

Assumption 1.1 (Fixed-design Gaussian regression). We observe $D_n = \{(x_i, y_i)\}_{i=1}^n$ where the design points $x_1, \dots, x_n$ are the fixed regular grid $\{(\frac{i_1-1/2}{n^{1/d}}, \dots, \frac{i_d-1/2}{n^{1/d}}) : i_\ell \in [n^{1/d}]\}$ in $[0, 1]^d$ (so $n^{1/d} \in \mathbb{N}$), and

$$y_i = f^*(x_i) + \xi_i, \quad \xi_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad i \in [n], \quad (2)$$

for a known variance $\sigma^2 > 0$ and an unknown regression function $f^* \in \mathcal{W}$. We write P_f for the law of D_n under regression function f .

Remark 1.2. The grid regularity is used only through Lemma 4.1, where it lets us compare the empirical sum $\frac{1}{n} \sum_i u(x_i)^2$ to $\|u\|_{L^2}^2$ for the localized perturbations built in Definition 1.4; any design whose cells each contain a comparable number of points would serve equally well. The Gaussian-noise and known- σ^2 choices are what make the pairwise Kullback–Leibler divergence in Lemma 4.1 a closed-form mean-shift expression.

Fact 1.3 (Gaussian fixed-design KL). Under Assumption 1.1, for any two regression functions f, h ,

$$\text{KL}(P_f \| P_h) = \frac{1}{2\sigma^2} \sum_{i=1}^n (f(x_i) - h(x_i))^2. \quad (3)$$

Proof. Under P_f the observations are independent with $y_i \sim \mathcal{N}(f(x_i), \sigma^2)$, and likewise under P_h ; the designs coincide. Hence

$$\begin{aligned} \text{KL}(P_f \| P_h) &= \sum_{i=1}^n \text{KL}(\mathcal{N}(f(x_i), \sigma^2) \| \mathcal{N}(h(x_i), \sigma^2)) \\ &= \sum_{i=1}^n \frac{(f(x_i) - h(x_i))^2}{2\sigma^2}, \end{aligned}$$

where the first equality is the tensorization of KL over the independent coordinates and the second is the closed form for the KL between two univariate Gaussians of common variance σ^2 . This gives the assertion. $\square$

Definition 1.4 (Localized bump family). Fix once and for all a function $g \in C_c^\infty((0,1)^d)$ with $g \not\equiv 0$, and set $c_g := \|g\|_{L^2}^2 > 0$ and $C_g := \sum_{|\alpha| \leq s} \|\partial^\alpha g\|_{L^2}^2$. For an integer resolution $m \geq 2$, partition $[0,1]^d$ into the $M_0 := m^d$ subcubes of side $1/m$, and let $x^{(j)} \in [0,1]^d$, $j \in [M_0]$, be the "lower corners" so that $x \mapsto m(x - x^{(j)})$ maps the j -th subcube onto $(0,1)^d$. For an amplitude $\omega > 0$ and a codeword $\tau \in \{0,1\}^{M_0}$, define

$$u_\tau(x) := \sum_{j=1}^{M_0} \tau_j \frac{\omega}{m^s} g\left(m(x - x^{(j)})\right), \quad x \in [0,1]^d. \quad (4)$$

The summands have pairwise disjoint supports (one per subcube).

2 The Varshamov–Gilbert packing

The first ingredient is a combinatorial packing of the binary hypercube: a large set of codewords that are pairwise far apart in Hamming distance. This is the classical Varshamov–Gilbert bound, which we quote.

Lemma 2.1 (Varshamov–Gilbert, [1, Lemma 2.9]). *Let $M_0 \geq 8$. There exists a set of codewords $\{\tau^{(1)}, \dots, \tau^{(M)}\} \subseteq \{0,1\}^{M_0}$ with*

$$M \geq 2^{M_0/8} \quad \text{and} \quad \rho_H\left(\tau^{(k)}, \tau^{(k')}\right) \geq \frac{M_0}{8} \quad \text{for all } 1 \leq k < k' \leq M. \quad (5)$$

Remark 2.2. We apply Lemma 2.1 with $M_0 = m^d$, which satisfies $M_0 \geq 8$ for every $m \geq 2$ when $d \geq 1$ (and for $m \geq 3$ when $d = 1$; we take m large in the end, so this is harmless). The two conclusions feed the two sides of the Fano bound: the Hamming separation $\rho_H \geq M_0/8$ becomes an L^2 separation of the hypothesis family in Lemma 3.1, while the cardinality $\log M \geq \frac{M_0}{8} \log 2$ becomes the entropy term $\log M$ in the denominator of the Fano inequality Eq. (11) used in Theorem 6.1.

3 The hypothesis family: feasibility and separation

We now turn the codewords of Lemma 2.1 into a finite family of regression functions inside the Sobolev ball $\mathcal{W}$ that are pairwise separated in L^2 . The amplitude ω is the single free parameter; it is tuned in Theorem 6.1.

Lemma 3.1 (Feasibility and L^2 separation). *Let g, c_g, C_g and u_τ be as in Definition 1.4, and let $\{\tau^{(1)}, \dots, \tau^{(M)}\}$ be the codewords of Lemma 2.1 with $M_0 = m^d$. Then for every codeword,*

$$\|u_{\tau^{(k)}}\|_{W_2^s}^2 \leq C_g \omega^2, \quad (6)$$

so that $\omega \leq \omega_0 := C_g^{-1/2}$ implies $u_{\tau^{(k)}} \in \mathcal{W}$ for all k ; and for every pair $k \neq k'$,

$$\|u_{\tau^{(k)}} - u_{\tau^{(k')}}\|_{L^2}^2 \geq \frac{c_g}{8} \frac{\omega^2}{m^{2s}}. \quad (7)$$

Proof. The proof is two change-of-variables computations: the first bounds the Sobolev norm of a single bump (giving feasibility after summing disjoint cells), the second turns the Hamming separation of Lemma 2.1 into the L^2 separation Eq. (7). Throughout, fix the substitution $y = m(x - x^{(j)})$ on the j -th subcube, under which $dx = m^{-d} dy$ and y ranges over $(0, 1)^d$.

Step 1: Sobolev norm of one bump. Fix a cell j and a multi-index α with $|\alpha| = r \leq s$. Since $\partial^\alpha [g(m(x - x^{(j)}))] = m^r (\partial^\alpha g)(m(x - x^{(j)}))$,

$$\begin{aligned} \int_{[0,1]^d} \left| \partial^\alpha \left(\frac{\omega}{m^s} g(m(x - x^{(j)})) \right) \right|^2 dx &= \frac{\omega^2}{m^{2s}} m^{2r} \int_{[0,1]^d} |(\partial^\alpha g)(m(x - x^{(j)}))|^2 dx \\ &= \frac{\omega^2}{m^{2s}} m^{2r} m^{-d} \int_{(0,1)^d} |(\partial^\alpha g)(y)|^2 dy \\ &= \frac{\omega^2}{m^{2(s-r)}} m^{-d} \|\partial^\alpha g\|_{L^2}^2 \\ &\leq \omega^2 m^{-d} \|\partial^\alpha g\|_{L^2}^2, \end{aligned}$$

where the first step differentiates the rescaled bump and pulls out the factor m^r , the second step applies the change of variables $y = m(x - x^{(j)})$ together with $\text{supp } g \subseteq (0, 1)^d$, the third step collects powers of m , and the last step uses $r \leq s$ so that $m^{-2(s-r)} \leq 1$.

Step 2: feasibility, Eq. (6). The M_0 summands in $u_{\tau^{(k)}}$ have disjoint supports, so all their mixed partials are orthogonal in L^2 and the squared norms add. Hence

$$\begin{aligned} \|u_{\tau^{(k)}}\|_{W_2^s}^2 &= \sum_{|\alpha| \leq s} \sum_{j=1}^{M_0} \tau_j^{(k)} \int_{[0,1]^d} \left| \partial^\alpha \left(\frac{\omega}{m^s} g(m(x - x^{(j)})) \right) \right|^2 dx \\ &\leq \sum_{|\alpha| \leq s} \left\| \tau^{(k)} \right\|_0 \omega^2 m^{-d} \|\partial^\alpha g\|_{L^2}^2 \\ &\leq \omega^2 m^{-d} M_0 \sum_{|\alpha| \leq s} \|\partial^\alpha g\|_{L^2}^2 \\ &= \omega^2 C_g, \end{aligned}$$

where the first step expands the definitions Eq. (1) and Eq. (4) and uses disjoint-support orthogonality, the second step inserts the per-bump bound from Step 1, the third step bounds the Hamming weight $\|\tau^{(k)}\|_0 \leq M_0$, and the last step uses $M_0 = m^d$ and the definition $C_g = \sum_{|\alpha| \leq s} \|\partial^\alpha g\|_{L^2}^2$. This proves Eq. (6); choosing $\omega \leq \omega_0 = C_g^{-1/2}$ gives $\|u_{\tau^{(k)}}\|_{W_2^s} \leq 1$, i.e. $u_{\tau^{(k)}} \in \mathcal{W}$.

Step 3: L^2 separation, Eq. (7). For one cell j , the same change of variables (with $r = 0$) gives the L^2 mass of a single bump,

$$\int_{[0,1]^d} \left| \frac{\omega}{m^s} g(m(x - x^{(j)})) \right|^2 dx = \frac{\omega^2}{m^{2s}} m^{-d} \|g\|_{L^2}^2 = \frac{c_g \omega^2}{m^{2s+d}}. \quad (8)$$

The difference $u_{\tau^{(k)}} - u_{\tau^{(k')}}$ is supported on exactly the cells where $\tau_j^{(k)} \neq \tau_j^{(k')}$, on each of which it equals $\pm \frac{\omega}{m^s} g(m(x - x^{(j)}))$. By disjointness of supports,

$$\begin{aligned} \|u_{\tau^{(k)}} - u_{\tau^{(k')}}\|_{L^2}^2 &= \sum_{j=1}^{M_0} \mathbf{1}\{\tau_j^{(k)} \neq \tau_j^{(k')}\} \int_{[0,1]^d} \left| \frac{\omega}{m^s} g(m(x - x^{(j)})) \right|^2 dx \\ &= \rho_H(\tau^{(k)}, \tau^{(k')}) \cdot \frac{c_g \omega^2}{m^{2s+d}} \\ &\geq \frac{M_0}{8} \cdot \frac{c_g \omega^2}{m^{2s+d}} \\ &= \frac{c_g}{8} \frac{\omega^2}{m^{2s}}, \end{aligned}$$

where the first step uses disjoint-support additivity of the L^2 norm, the second step inserts the per-cell mass Eq. (8) and recognizes the number of differing coordinates as the Hamming distance, the third step applies the Varshamov–Gilbert separation $\rho_H \geq M_0/8$ from Eq. (5), and the last step uses $M_0 = m^d$. This is Eq. (7), completing the proof. $\square$

4 Pairwise Kullback–Leibler bound

To control the Fano numerator we bound the pairwise KL divergence between the data laws induced by two hypotheses. The fixed-design grid enters here, and only here, through a counting bound on the number of design points per cell.

Lemma 4.1 (Pairwise KL bound). *Assume in addition that m divides $n^{1/d}$, so that each of the $M_0 = m^d$ subcubes of Definition 1.4 contains exactly n/M_0 design points. Let $G_\infty := \|g\|_\infty^2$. Then for all $k \neq k'$,*

$$\text{KL}(P_{u_{\tau^{(k)}}} \parallel P_{u_{\tau^{(k')}}}) \leq \frac{G_\infty}{2\sigma^2} \frac{n\omega^2}{m^{2s}}. \quad (9)$$

Proof. Write $\Delta u := u_{\tau^{(k)}} - u_{\tau^{(k')}}$ for brevity. By Fact 1.3,

$$\text{KL}(P_{u_{\tau^{(k)}}} \parallel P_{u_{\tau^{(k')}}}) = \frac{1}{2\sigma^2} \sum_{i=1}^n (\Delta u(x_i))^2, \quad (10)$$

so it suffices to bound the design sum $\sum_i (\Delta u(x_i))^2$. Since Δu is supported on the cells where $\tau^{(k)}$ and $\tau^{(k')}$ differ and on each such cell j equals $\pm \frac{\omega}{m^s} g(m(x - x^{(j)}))$, and since each cell carries exactly n/M_0 design points,

$$\begin{aligned} \sum_{i=1}^n (\Delta u(x_i))^2 &= \sum_{j: \tau_j^{(k)} \neq \tau_j^{(k')}} \sum_{i: x_i \in \text{cell } j} \left(\frac{\omega}{m^s} g(m(x_i - x^{(j)})) \right)^2 \\ &\leq \sum_{j: \tau_j^{(k)} \neq \tau_j^{(k')}} \frac{n}{M_0} \cdot \frac{\omega^2}{m^{2s}} G_\infty \\ &= \rho_H(\tau^{(k)}, \tau^{(k')}) \cdot \frac{n}{M_0} \cdot \frac{\omega^2}{m^{2s}} G_\infty \\ &\leq M_0 \cdot \frac{n}{M_0} \cdot \frac{\omega^2}{m^{2s}} G_\infty = G_\infty \frac{n\omega^2}{m^{2s}}, \end{aligned}$$

where the first step partitions the sum over the differing cells and uses the explicit form of Δu there, the second step bounds each summand by its maximum $\frac{\omega^2}{m^{2s}} G_\infty$ over the n/M_0 points in the cell, the third step recognizes the number of differing cells as the Hamming distance, and the last step bounds $\rho_H \leq M_0$ and cancels M_0 . Inserting this into Eq. (10) yields Eq. (9), as desired. $\square$

5 The local Fano inequality

The final ingredient is the information-theoretic core: Fano's inequality in the form that lower-bounds the minimax testing error by the entropy $\log M$ minus the mutual information, together with the standard bound of the mutual information by the pairwise KL diameter.

Lemma 5.1 (Local Fano method, [1, Theorem 2.5 and Corollary 2.6]). *Let $M \geq 2$ and let V be uniform on $[M]$. Suppose that conditionally on $V = k$ the observation Y has law P_k . Then for every (possibly randomized) test $\psi(Y)$ taking values in $[M]$,*

$$\inf_{\psi} \mathbb{P}[\psi(Y) \neq V] \geq 1 - \frac{I(V; Y) + \log 2}{\log M}, \quad I(V; Y) \leq \frac{1}{M^2} \sum_{k, k'} \text{KL}(P_k \| P_{k'}), \quad (11)$$

where $I(V; Y)$ is the mutual information between V and Y . In particular, $I(V; Y) \leq \max_{k \neq k'} \text{KL}(P_k \| P_{k'})$ by bounding the average by the maximum.

Remark 5.2. The two displays in Eq. (11) are the Fano testing inequality and the convexity bound $I(V; Y) \leq \frac{1}{M^2} \sum_{k, k'} \text{KL}(P_k \| P_{k'}) \leq \max_{k \neq k'} \text{KL}(P_k \| P_{k'})$, respectively. In Theorem 6.1 we apply Lemma 5.1 with $P_k = P_{u_{\tau(k)}}$ and M the packing cardinality of Lemma 2.1; the mutual-information bound is supplied by Lemma 4.1.

6 The minimax lower bound

We now assemble Lemmas 2.1, 3.1, 4.1 and 5.1 into the minimax lower bound, choosing the resolution m and amplitude ω to balance the L^2 separation against the KL diameter.

Theorem 6.1 (Sobolev minimax lower bound). *Consider fixed-design Gaussian regression (Assumption 1.1) over the Sobolev ball $\mathcal{W} = \{f : \|f\|_{W_2^s} \leq 1\}$ on $[0, 1]^d$ with integer smoothness $s \geq 1$. There is a constant $c = c(s, d, g, \sigma^2) > 0$ such that, for all n large enough,*

$$\inf_{\hat{f}} \sup_{f^* \in \mathcal{W}} \mathbb{E}_{D_n} \left\| \hat{f} - f^* \right\|_{L^2}^2 \geq c n^{-2s/(2s+d)}, \quad (12)$$

the infimum running over all estimators $\hat{f} = \hat{f}(D_n)$.

Proof. The proof reduces estimation to an M -ary hypothesis test over the bump family of Definition 1.4, then balances the L^2 separation against the information content. Fix the bump g and the constants c_g, C_g, G_∞ together with $\omega_0 := C_g^{-1/2}$ from Definition 1.4 and Lemmas 3.1 and 4.1, and set the amplitude to its largest feasible value $\omega = \omega_0$ throughout.

Choice of resolution. Define the constant and the resolution

$$B := \left(\frac{64 G_\infty \omega_0^2}{\sigma^2 \log 2} \right)^{1/(2s+d)}, \quad m := \left\lceil B n^{1/(2s+d)} \right\rceil, \quad (13)$$

taken along the subsequence of sample sizes for which $m \mid n^{1/d}$ (so Lemma 4.1 applies). By construction $m \geq B n^{1/(2s+d)}$, and for n large enough $m \leq 2B n^{1/(2s+d)}$, so

$$B n^{1/(2s+d)} \leq m \leq 2B n^{1/(2s+d)}. \quad (14)$$

Step 1: the hypotheses are feasible and 2Δ -separated. Let $\{\tau^{(1)}, \dots, \tau^{(M)}\}$ be the Varshamov–Gilbert codewords of Lemma 2.1 with $M_0 = m^d \geq 8$, and set $f_k := u_{\tau^{(k)}}$. By Lemma 3.1 with $\omega = \omega_0$, every $f_k \in \mathcal{W}$, and for $k \neq k'$,

$$\|f_k - f_{k'}\|_{L^2}^2 \geq \frac{c_g}{8} \frac{\omega_0^2}{m^{2s}} =: (2\Delta)^2, \quad \text{so} \quad \Delta^2 = \frac{c_g \omega_0^2}{32} \frac{1}{m^{2s}}. \quad (15)$$

Moreover, by Eq. (5) the cardinality satisfies

$$\log M \geq \frac{m^d}{8} \log 2. \quad (16)$$

Step 2: the KL diameter is at most $\frac{1}{4} \log M$. By Lemma 4.1 with $\omega = \omega_0$, and then the resolution lower bound $m \geq B n^{1/(2s+d)}$ from Eq. (14),

$$\begin{aligned} \max_{k \neq k'} \text{KL}(P_{f_k} \parallel P_{f_{k'}}) &\leq \frac{G_\infty}{2\sigma^2} \frac{n \omega_0^2}{m^{2s}} \\ &= \frac{G_\infty \omega_0^2}{2\sigma^2} \frac{n}{m^{2s+d}} m^d \\ &\leq \frac{G_\infty \omega_0^2}{2\sigma^2} \frac{1}{B^{2s+d}} m^d \\ &= \frac{G_\infty \omega_0^2}{2\sigma^2} \cdot \frac{\sigma^2 \log 2}{64 G_\infty \omega_0^2} m^d = \frac{\log 2}{128} m^d \leq \frac{1}{4} \log M, \end{aligned}$$

where the first step is Lemma 4.1, the second step multiplies and divides by m^d , the third step uses $m^{2s+d} \geq B^{2s+d} n$ (hence $n/m^{2s+d} \leq B^{-(2s+d)}$) from Eq. (14), the fourth step inserts the value of B from Eq. (13) and cancels, and the last step uses $\frac{\log 2}{128} m^d \leq \frac{1}{4} \cdot \frac{m^d}{8} \log 2 \leq \frac{1}{4} \log M$, where the first inequality holds since $\frac{1}{128} \leq \frac{1}{4} \cdot \frac{1}{8} = \frac{1}{32}$ and the second is Eq. (16).

Step 3: Fano lower-bounds the testing error. Apply Lemma 5.1 with $P_k = P_{f_k}$ and V uniform on $[M]$ (valid since $M \geq 2^{M_0/8} \geq 2$). Combining its mutual-information bound with Step 2,

$$\begin{aligned} \inf_{\psi} \mathbb{P}[\psi(Y) \neq V] &\geq 1 - \frac{I(V; Y) + \log 2}{\log M} \\ &\geq 1 - \frac{\frac{1}{4} \log M + \log 2}{\log M} \\ &\geq 1 - \frac{1}{4} - \frac{\log 2}{\log M} \geq \frac{1}{2}, \end{aligned}$$

where the first step is the Fano inequality Eq. (11), the second step inserts $I(V; Y) \leq \max_{k \neq k'} \text{KL}(P_{f_k} \parallel P_{f_{k'}}) \leq \frac{1}{4} \log M$ from Eq. (11) and Step 2, the third step splits the fraction, and the last step uses $\log 2 / \log M \leq 1/4$: by Eq. (16) and $m^d \rightarrow \infty$ we have $\log M \geq \frac{m^d}{8} \log 2 \geq 4 \log 2$ for n large enough.

Step 4: from testing error to risk. Fix any estimator $\hat{f}$ and define the test $\psi(Y) := \arg \min_{k \in [M]} \|\hat{f} - f_k\|_{L^2}$ (ties broken arbitrarily). If $\psi(Y) \neq V = k$, then $f_\psi \neq f_k$ are two distinct hypotheses, and by the triangle inequality and the separation Eq. (15),

$$\begin{aligned} 2 \|\hat{f} - f_k\|_{L^2} &\geq \|\hat{f} - f_k\|_{L^2} + \|\hat{f} - f_\psi\|_{L^2} \\ &\geq \|f_k - f_\psi\|_{L^2} \geq 2\Delta, \end{aligned}$$

where the first step uses that ψ minimizes the distance to $\hat{f}$ (so $\|\hat{f} - f_\psi\|_{L^2} \leq \|\hat{f} - f_k\|_{L^2}$), the second step is the triangle inequality, and the third step is Eq. (15). Hence $\|\hat{f} - f_k\|_{L^2} \geq \Delta$ on the event $\{\psi \neq V\}$, so $\|\hat{f} - f_k\|_{L^2}^2 \geq \Delta^2 \mathbf{1}\{\psi \neq V\}$. Taking expectations and the supremum over the family,

$$\begin{aligned} \sup_{f^* \in \mathcal{W}} \mathbb{E}_{D_n} \|\hat{f} - f^*\|_{L^2}^2 &\geq \max_{k \in [M]} \mathbb{E}_{P_{f_k}} \|\hat{f} - f_k\|_{L^2}^2 \\ &\geq \frac{1}{M} \sum_{k=1}^M \mathbb{E}_{P_{f_k}} \|\hat{f} - f_k\|_{L^2}^2 \\ &\geq \Delta^2 \cdot \frac{1}{M} \sum_{k=1}^M \mathbb{P}_{P_{f_k}}[\psi(Y) \neq k] \\ &= \Delta^2 \cdot \mathbb{P}[\psi(Y) \neq V] \geq \frac{\Delta^2}{2}, \end{aligned}$$

where the first step restricts the supremum to the finite family $\{f_k\}$ (each in $\mathcal{W}$ by Step 1), the second step bounds a maximum below by an average, the third step inserts $\|\hat{f} - f_k\|_{L^2}^2 \geq \Delta^2 \mathbf{1}\{\psi \neq k\}$, the fourth step recognizes the average over the uniform prior as $\mathbb{P}[\psi(Y) \neq V]$, and the last step uses the Fano bound of Step 3. As $\hat{f}$ was arbitrary, the same lower bound holds for $\inf_{\hat{f}}$.

Step 5: the rate. Inserting Δ^2 from Eq. (15) and the upper bound $m \leq 2B n^{1/(2s+d)}$ from Eq. (14),

$$\begin{aligned} \inf_{\hat{f}} \sup_{f^* \in \mathcal{W}} \mathbb{E}_{D_n} \|\hat{f} - f^*\|_{L^2}^2 &\geq \frac{\Delta^2}{2} = \frac{c_g \omega_0^2}{64} \frac{1}{m^{2s}} \\ &\geq \frac{c_g \omega_0^2}{64} \frac{1}{(2B)^{2s}} n^{-2s/(2s+d)} = c n^{-2s/(2s+d)}, \end{aligned}$$

where the first step is Step 4 and Eq. (15) (so $\frac{\Delta^2}{2} = \frac{1}{2} \cdot \frac{c_g \omega_0^2}{32 m^{2s}} = \frac{c_g \omega_0^2}{64 m^{2s}}$), the second step inserts $m \leq 2B n^{1/(2s+d)}$ hence $m^{-2s} \geq (2B)^{-2s} n^{-2s/(2s+d)}$, and the last step collects the (s, d, g, σ^2) -dependent factors into $c := \frac{c_g \omega_0^2}{64 (2B)^{2s}} > 0$. This is Eq. (12), completing the proof. $\square$

References

- [1] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer, New York, 2009. Varshamov–Gilbert bound: Lemma 2.9; Fano-type minimax lower bound and reduction to testing: Theorem 2.5 / Corollary 2.6.